Simultaneous quantification of ADHD drugs and endogenous DPPC in human breath reveals breath-blood correlations and suggests vapor-phase analyte collection

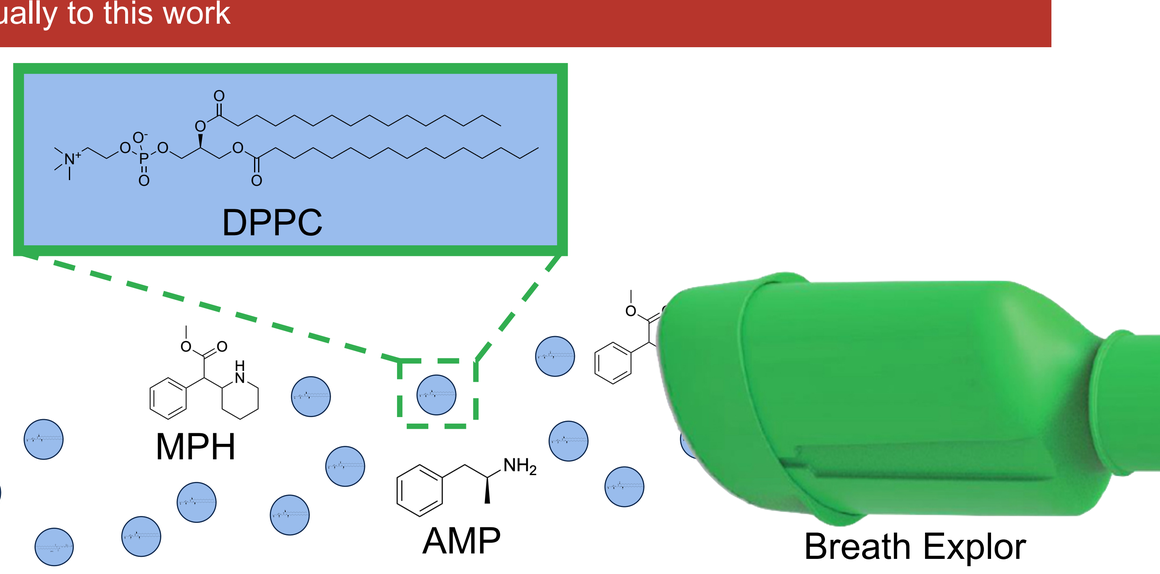
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1 Introduction

Exhaled breath contains compounds both in the gas phase and in aerosol particles derived from respiratory tract lining fluid (RTLF). Impaction-based devices such as the commercial “Breath Explor” were developed to sample aerosol particles and have been used to detect endogenous and exogenous compounds in breath. However, the exact collection mechanism of semi-volatile compounds such as amphetamine (AMP) and methylphenidate (MPH) is not fully understood.

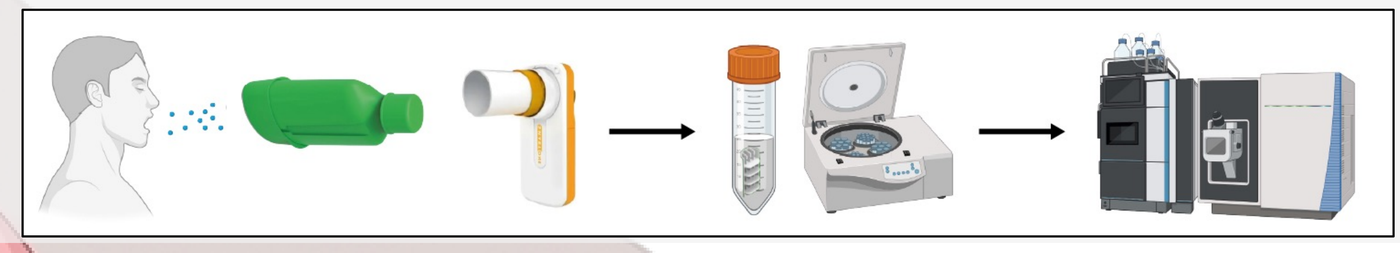


2 Method Overview

Breath and paired blood samples were collected from 18 adults receiving prescribed ADHD medication. Breath samples were collected from 12 deep exhalations using the impaction-based breath sampler Breath Explor. One cartridge from the sampler was extracted with 2 mL methanol (including 0.1% formic acid and labeled internal standard). The sample was up-concentrated in a vacuum centrifuge and dissolved in 60 µL of methanol and analyzed using a hydrophilic interaction liquid chromatography tandem mass spectrometry (HILIC-LC-MS/MS) method. Dipalmitoyl phosphatidylcholine (DPPC), the major phospholipid inside the RTLF, was measured simultaneously with AMP and MPH as a proxy for the amount of aerosol collected[1].

Figure 1 (left): Schematic overview of the analytical workflow.

Mobile phase	H_2O – ACNL, 4 mM NH_4FA, 0.1% FA
Column	BEH amide 1.7µm 2.1 x 150 mm
LOQ	AMP: 24 pg / c., MPH: 3.2 pg / c., DPPC 138 pg / c.



3 Results and Discussion

Large inter-individual variation in the mass of collected DPPC was observed (Figure 2), indicating substantial variation in the mass of aerosol particles sampled. Despite this variation, a correlation was observed between the amount of AMP and MPH collected in breath and their respective blood concentrations (Figure 3). Normalization by DPPC decreases this correlation, indicating that the sampling of both compounds is independent from the collection of aerosol particles and is probably related to adsorption.

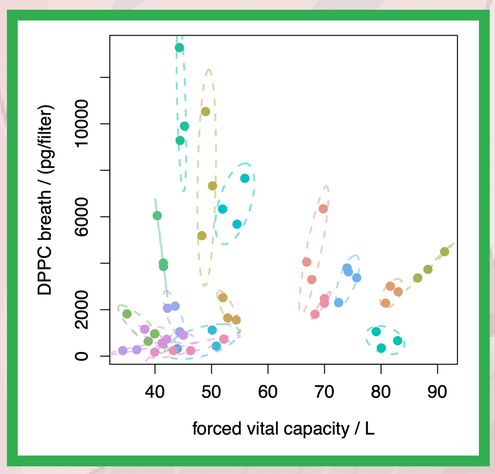
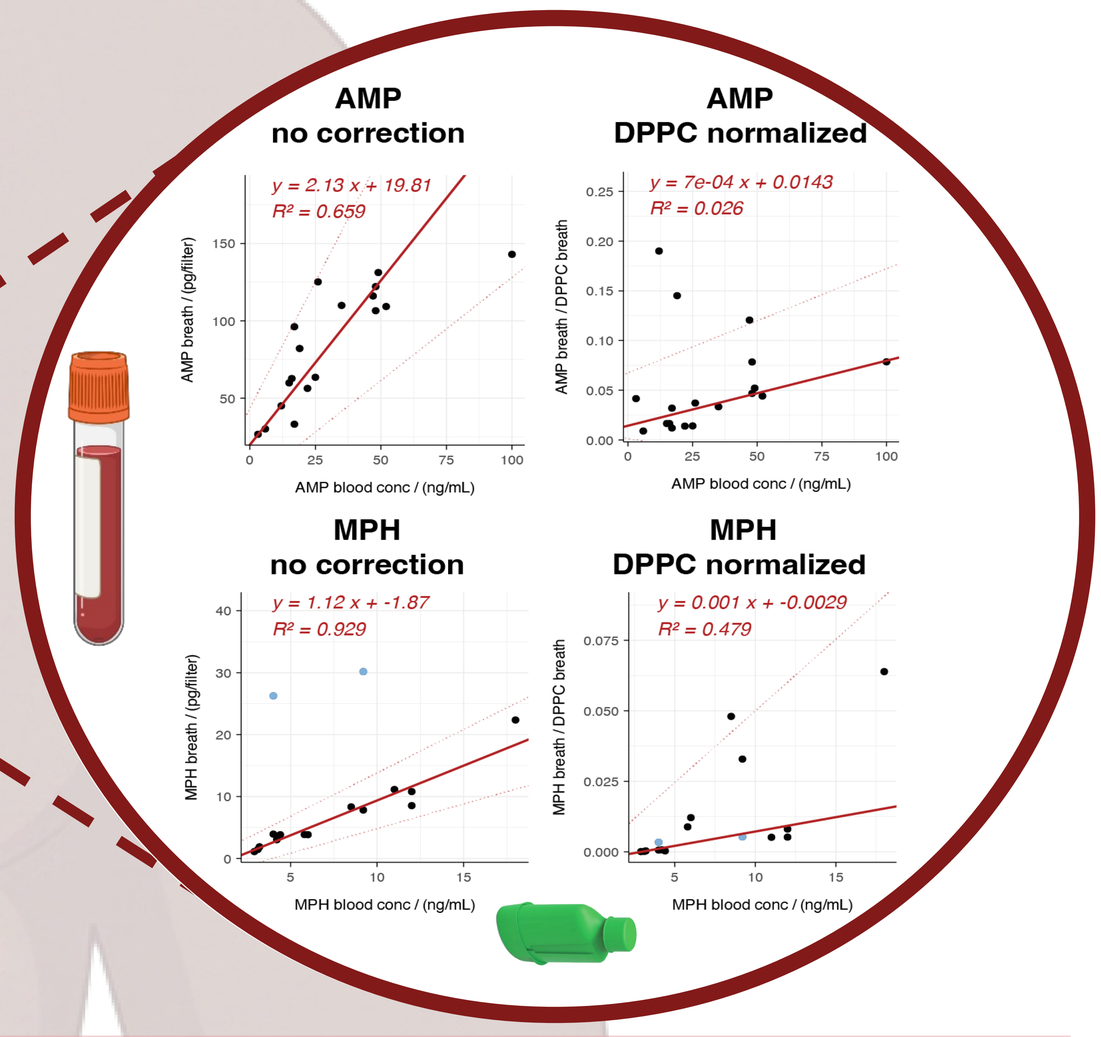


Figure 2 (left): Relationship between the amount of DPPC sampled and the total breath volume sampled (forced vital capacity). Measurements from the same person are shown in the same colour.

Figure 3 (right): Correlation between the amounts of AMP and MPH in breath and their concentrations in blood. Plots on the right side are normalized by the amount of DPPC in breath.



4 Conclusion

- There is a correlation between the amount of amphetamine and methylphenidate in breath and their blood concentrations.
- The trapping of both substances is independent of aerosol collection. They are present in the gas phase of breath and are probably adsorbed onto the filter.

References

[1] Respiratory Physiology & Neurobiology 243 (2017) 39–46

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